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NFC/RFID contactless readers: choose the right device for the right application

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Q&A session



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Solutions for NFC/RFID tags & readers

ST25 simply more connected

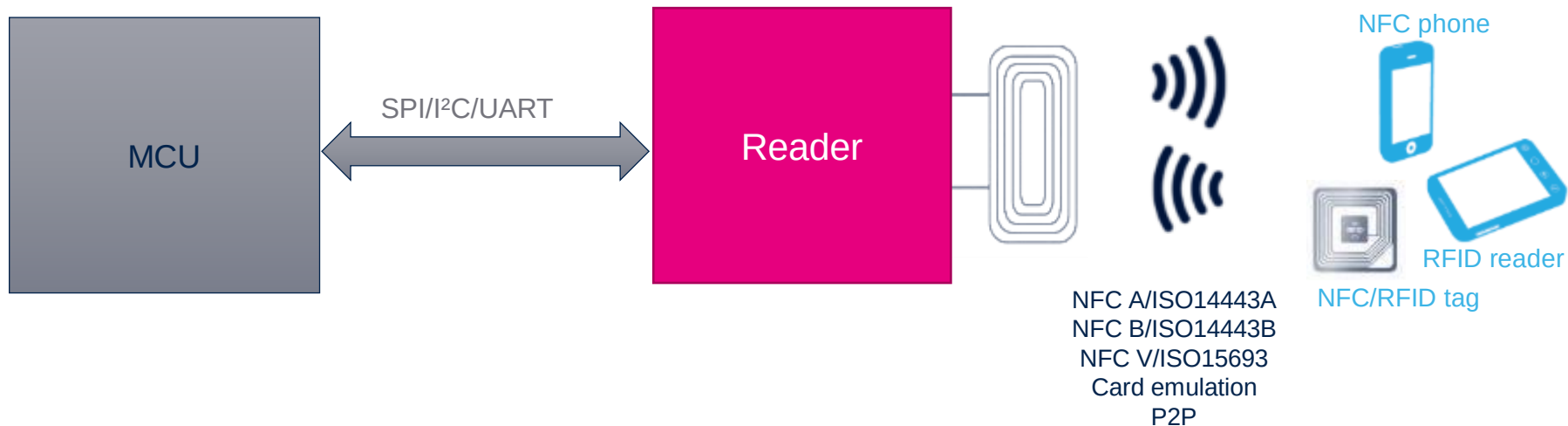


NFC technology at a glance

An interactive technology enabling engagement with IoT devices



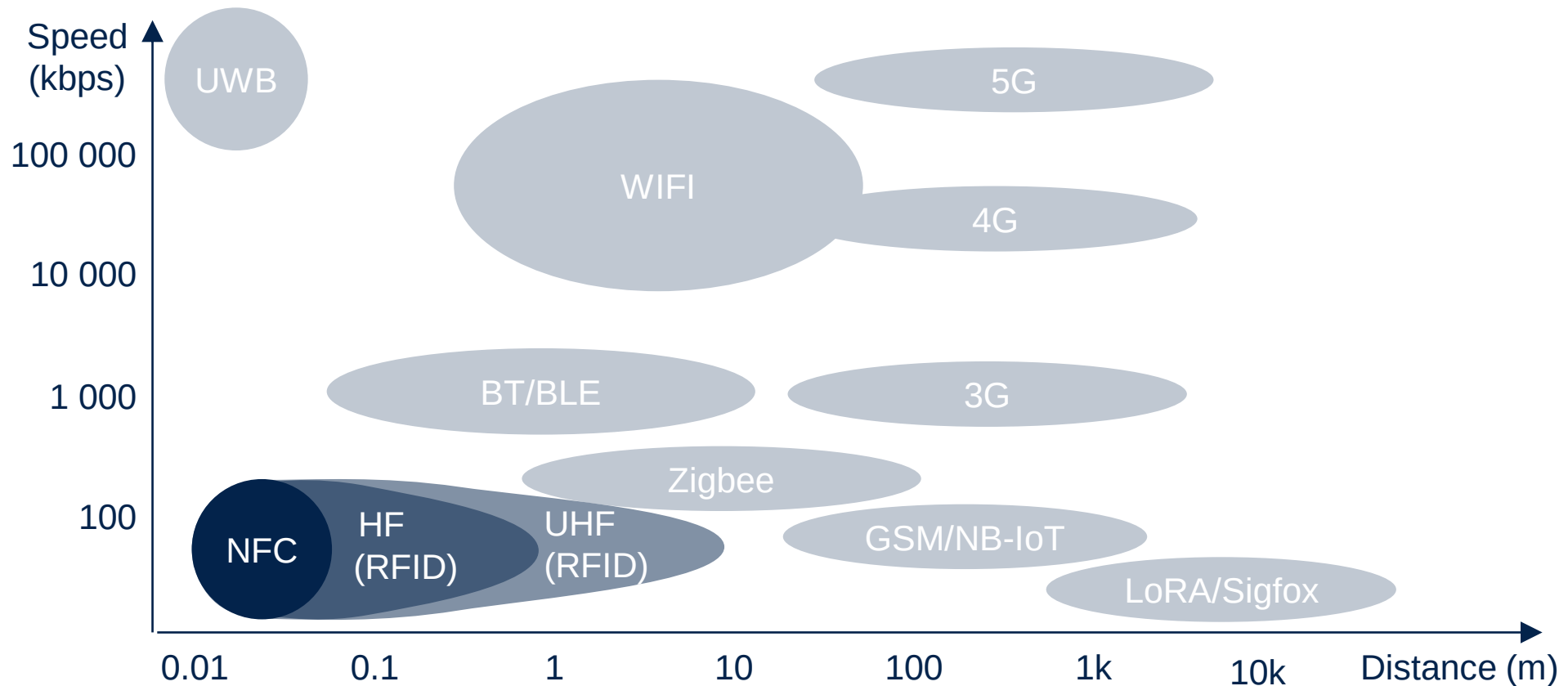
- Near Field Communication, a **short-range** wireless technology
 - Operating at **13.56MHz**
 - Based on the RFID HF standard (ISO14443 & ISO15693)
- **Interactive** and **zero power**, enabling convenient connection to the Internet of Things
- NFC is developed by the NFC Forum





NFC in the wireless spectrum

NFC is unique in the wireless spectrum: short distance, low data-rate & zero power consumption for the application

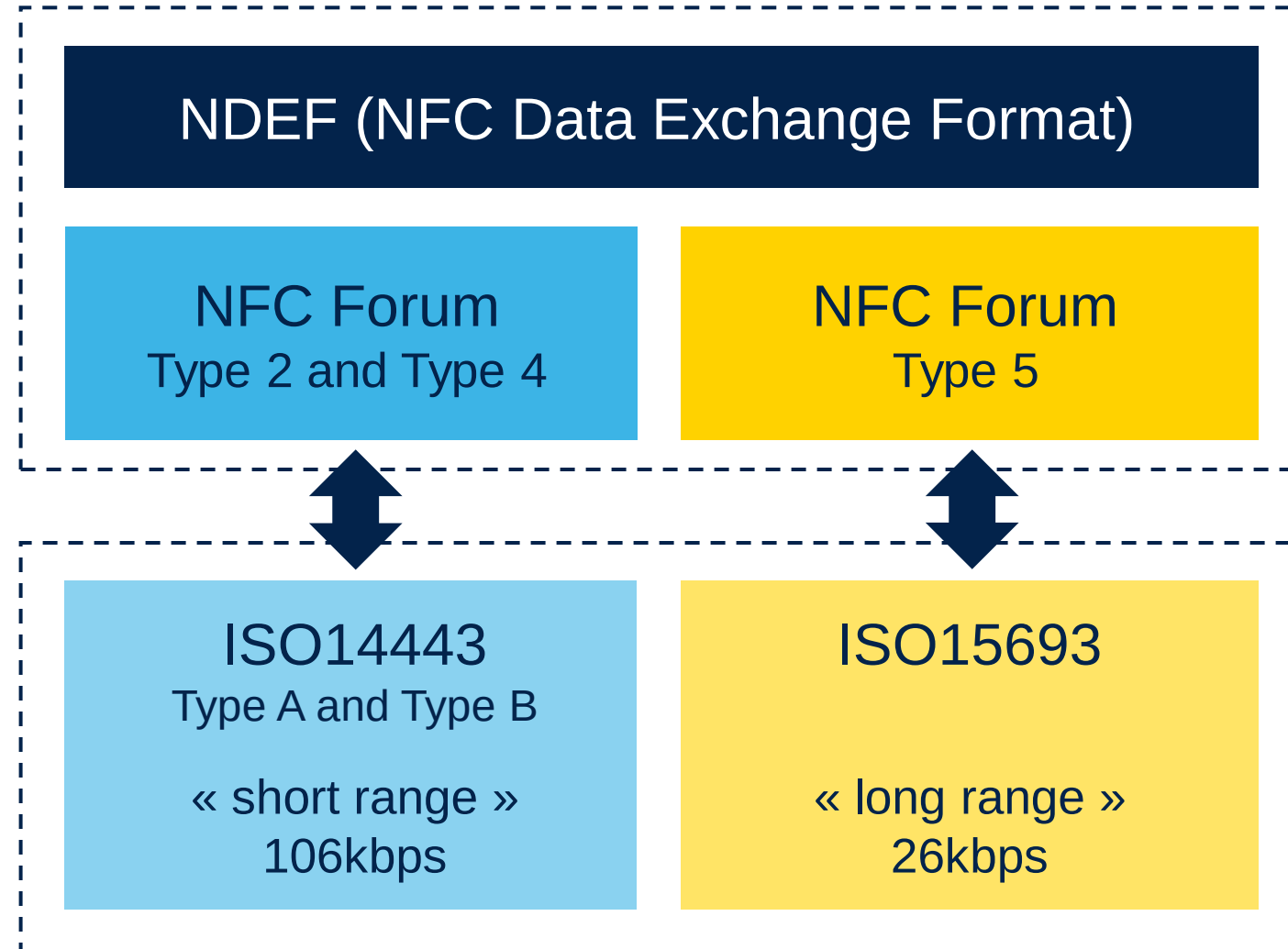


NFC Forum standards

NFC specification
→ upper layer SW

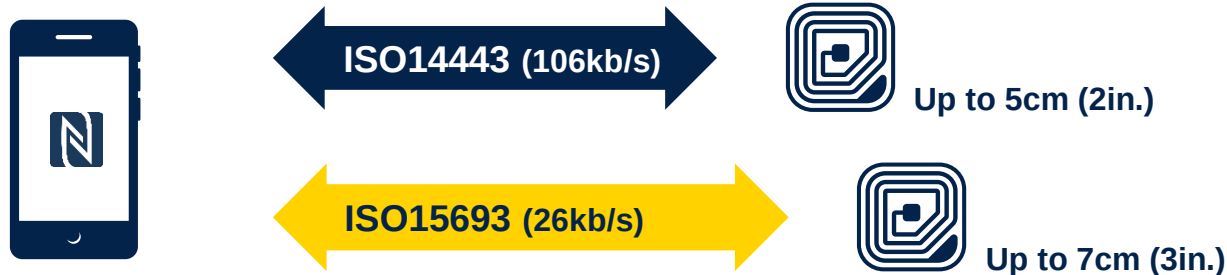


RFID HF ISO standards
→ HW/SW protocol



Typical NFC/RFID range

NFC phones



RFID readers



- ISO14443 (NFC Forum Type 2 & Type 4) is the **short range** standard while with higher RF speed
- **ISO15693 (NFC Forum Type 5) the long range standard**

ST25 products Families



ST25 product family

Consumer engagement, asset tracking, ticketing, brand protection, access control, gaming...

www.st.com/st25t

Tags



ST25T



13.56MHz



NFC phone/RFID reader

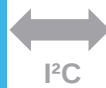
Industrial, lighting, metering, motor control, consumer, appliance, healthcare...

www.st.com/st25d

Dynamic tags



STM32
microcontroller



ST25D



13.56MHz



NFC phone/RFID reader

POS & mPOS terminals, automotive, access control, gaming, reader+tag...

www.st.com/st25r

Readers



STM32
microcontroller



ST25R



13.56MHz



NFC phone

or



or





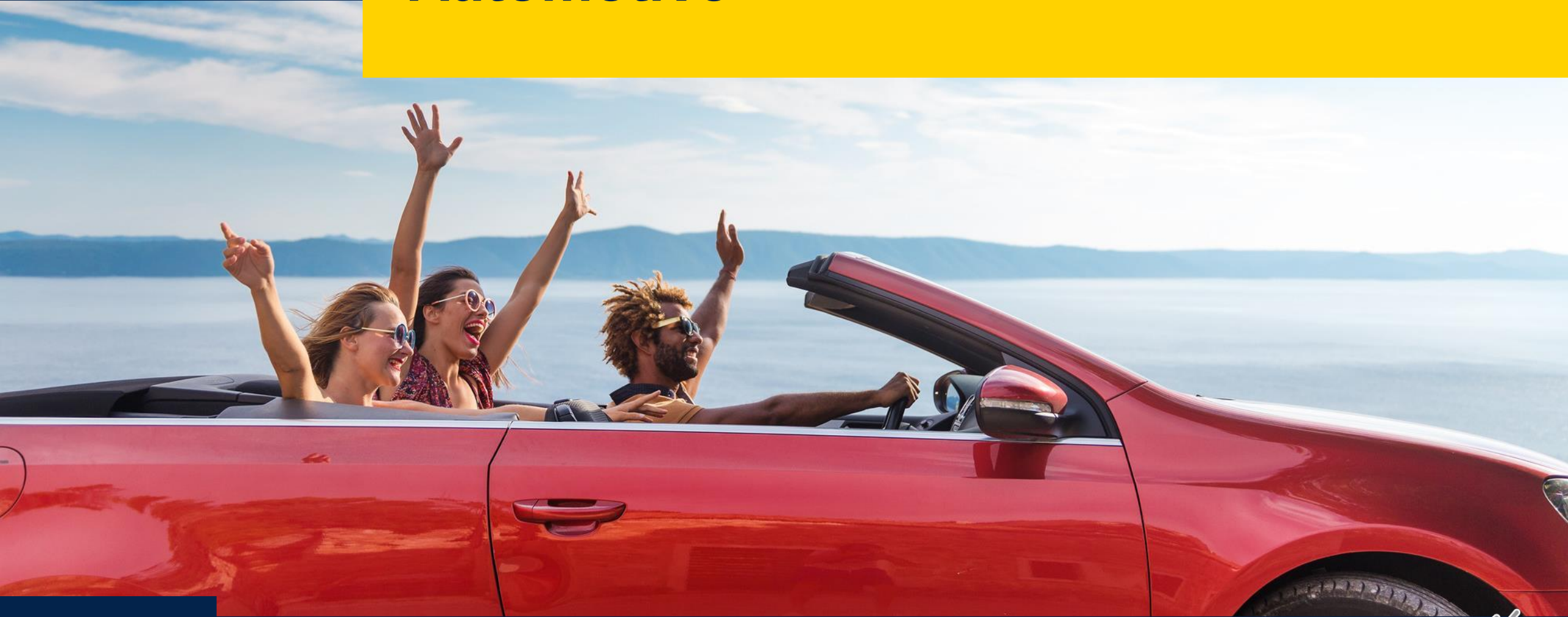
ST25R NFC/HF readers product family

	ST25R3912	ST25R3914/15	ST25R3916B/17B	ST25R3918	ST25R3920B
Description	Mid-Range NFC Forum reader	Automotive grade NFC Forum reader	High-performance NFC & EMVCo® reader	Multipurpose NFC transceiver	Automotive grade NFC Forum reader
Reader/Writer mode	ISO14443A/B ISO15693 FeliCa	ISO14443A/B ISO15693 FeliCa	ISO14443A/B ISO15693 FeliCa	ISO14443A/B ISO15693	ISO14443A/B ISO15693 FeliCa
Card emulation mode	-	-	Yes (16B)	Yes	Yes
AP2P mode	Initiator & Target	Initiator & Target	Initiator & Target (16B)	-	Initiator & Target
PP2P mode	Initiator	Initiator	Initiator & Target (16B)	Initiator & Target	Initiator & Target
RF speed	848kbps	848kbps	848kbps	848kbps	848kbps
Targeted market	Consumer, reader+tag	Automotive AEC-Q100 grade 1	Payment EMVCo® 3.1, Industrial, consumer	Reader+tag, IoT, gaming, consumer	Automotive AEC-Q100 grade 1
Advanced features	DPO, IWU	AAT (14), DPO, CIWU	AAT (16B), DPO, NSR, DSA, AWS, IWU, EMD	DPO, NSR, DSA, AWS, IWU, EMD	AAT, DPO, NSR, DSA, AWS, CIWU, EMD
Hardware interface	SPI 6Mbps	SPI 6Mbps	I²C & SPI 10Mbps	I²C & SPI 10Mbps	I²C & SPI 10Mbps
Power supply	2.4V–5.5V	2.4V–5.5V	2.4V–5.5V	2.4V–5.5V	2.4V–5.5V
Output power	1.0W	1.0W	1.6W	0.5W	1.6W
Temp range	-40°C to +125°C ^(J)	-40°C to +125°C ^(J)	-40°C to +105°C	-40°C to +85°C	-40°C to +125°C
Reader family	1 st generation		2 nd generation		
SW interface					

Applications & Use Cases



Automotive



Allow car access and ignition

- **NFC for car access and ignition**
 - Convenient and secure access to the car
 - Distribution of keys online to your NFC smartphone

ST solutions



• ST25R3920B

Secure wireless charging in central console

- **Secure wireless charging in the central console**
 - NFC reader monitors magnetic field from Qi charging
 - NFC reader avoids extreme high fields and its associated damage

ST solutions



• ST25R3920B

NFC for Payment



Payment Terminals

Payment made easy and secure



Payment & home banking

Full EMVCo® POS and mobile POS (mPOS)
For cashless transactions with minimal investment

Enhanced user experience

Quicker payment processing

ST solutions



- ST25R3916B
- ST25R3917B

Reader + Tag



Entertainment & Gaming

Enhance your gaming possibilities by using NFC



Figurines come to life,
Using NFC to connect to a gaming console

Player and figurine interaction
Without any external battery needs

ST solutions



- ST25R3912
- ST25R3918
- ST25 tag

Accessory Recognition

Brand recognition and parameters setting



NFC for accessory identification

Convenient and reliable brand identification

Brand recognition

Ensure that only branded accessories are used

ST solutions



- ST25R3912
- ST25R3918
- ST25 tag

Car After-Market Identification

Make sure you have the right spare part in your car



Car after-market identification

Accessory type recognition and authentication

ST solutions



- ST25R3914
- ST25R3915

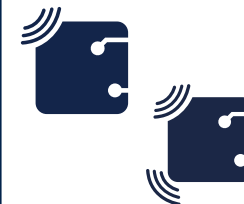
System Configuration

Get your device quickly and securely configured



Automatic system configuration
Upon accessory type recognition

ST solutions



- ST25R3912
- ST25R3918
- ST25D dynamic tag



ST25R NFC/HF readers product family

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SW interface					

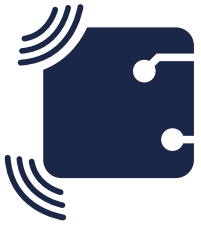


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1st Q&A Session

ST25 readers technical features





ST25R reader features

Several features and high performance for your product's success

Key features

- ✓ 1.6W highest output power
- ✓ Dynamic power control
- ✓ Noise suppression receivers
- ✓ Automatic antenna tuning
- ✓ AEC Q-100 qualified
- ✓ NFC Forum reader & universal device

Performances

- ✓ Up to 20cm (TN)/1m (TV) read range
- ✓ NFC Forum certified
- ✓ Low power modes



Interoperability



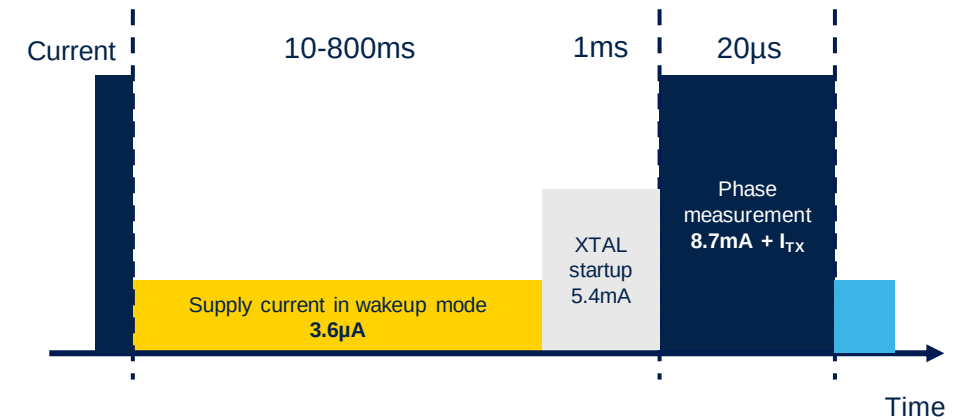
Wake-up methods



Low power wakeup

Low power wakeup will maximize your application lifetime

- The ST25R series offers an inductive low power card detection mode to reduce current consumption to a minimum while searching for tags or phones
 - Reduces average supply current
 - Ideal for battery powered applications like door locks, toys, automotive digital key...
 - Present at all ST25 NFC readers



- Fully programmable wakeup scheme.
- All relevant parameters like cycle time & sensitivity can be programmed and don't need MCU interaction.

Dynamic Power Output

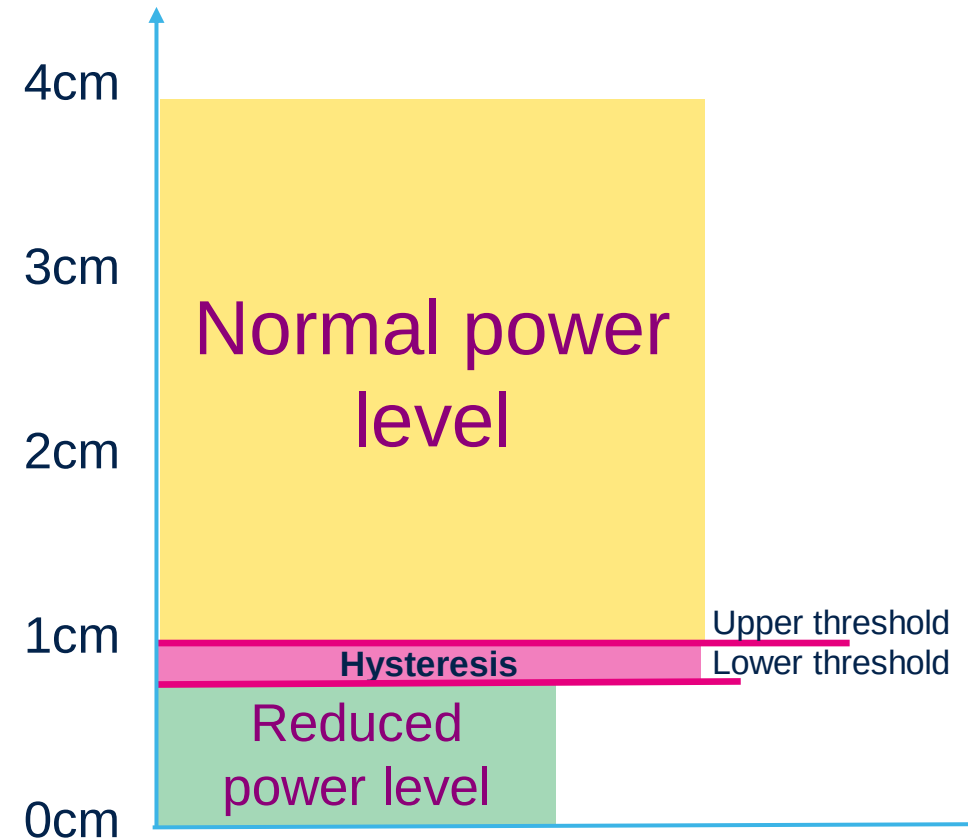




Dynamic power output (DPO)

DPO will keep power levels within limits of requirements

- Achieve min/max power limits easier
 - The DPO allows to adjust the output power dynamically via dynamic power output.
- Optimal performance from weak to strong card response
 - The DPO allows adaption to different power levels of card responses via active gain control.
- Improved noise immunity
 - The squelch feature allows scaling the signal level for improved noise immunity.

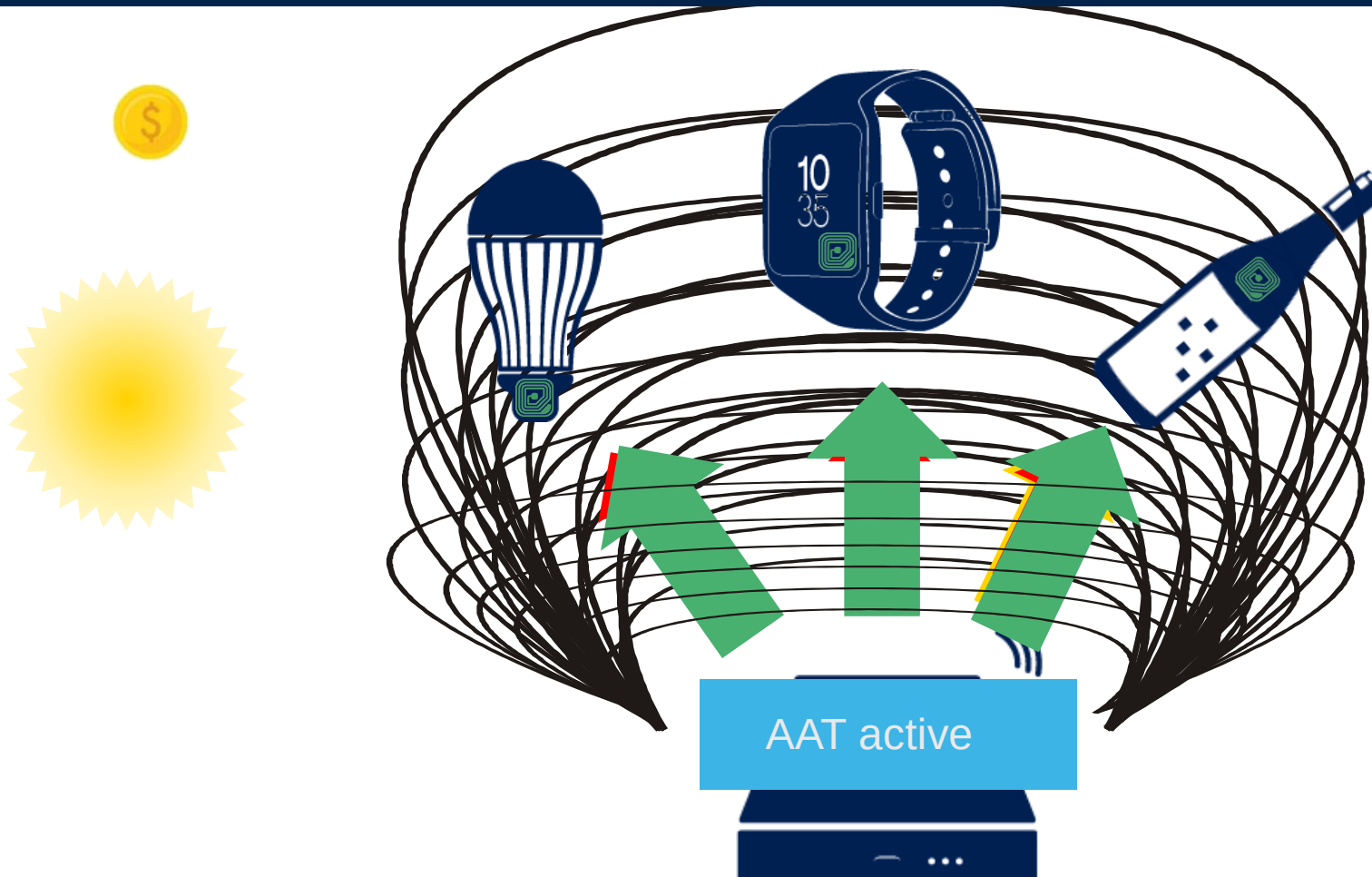


Automatic Antenna Tuning



Match the antenna well and make sure it stays tuned

Tags are detuning the antenna, reducing power transfer



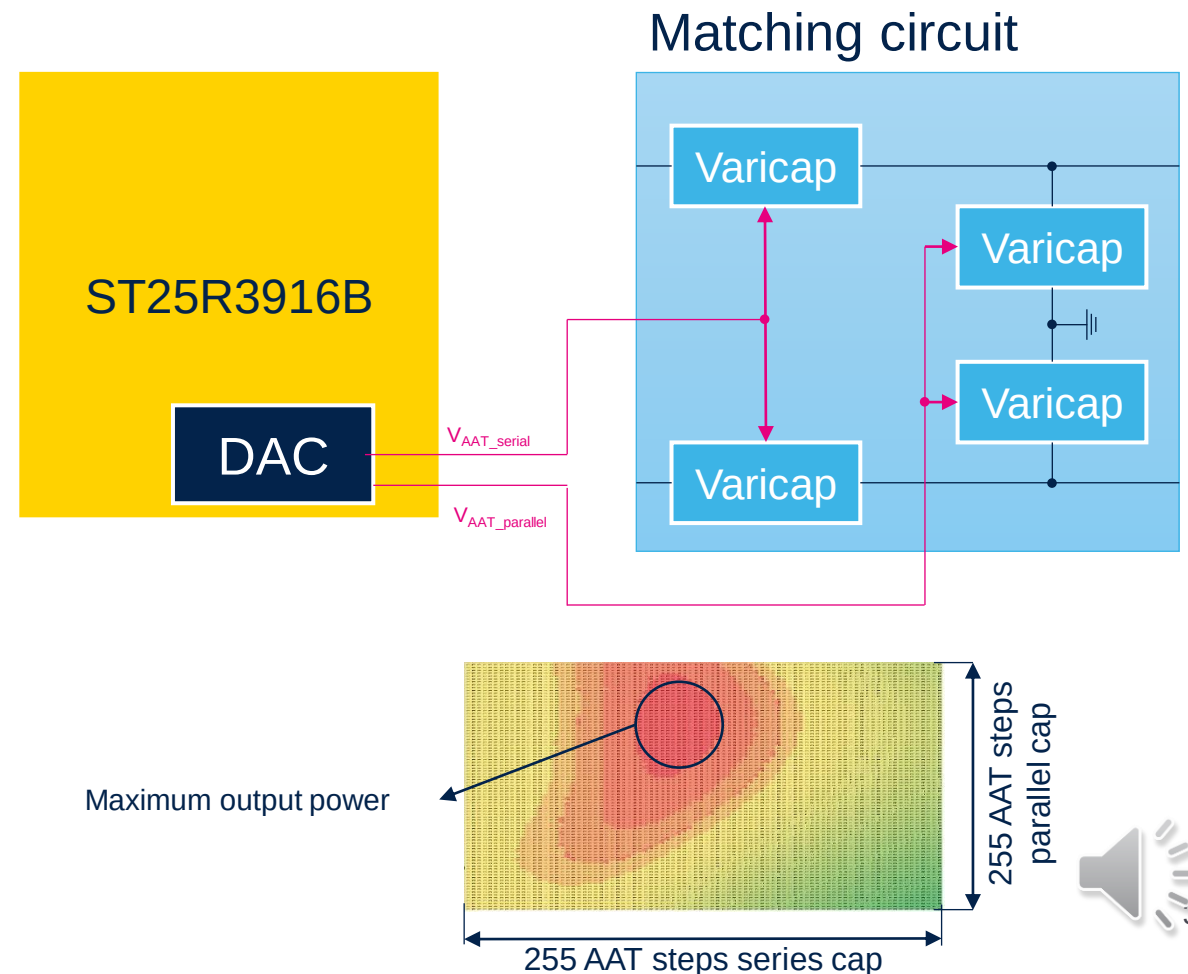
Adding multiple tags in the field detunes the field and drives less power into each tag.

Resulting in fails to detect/activate tags and having less power for device (energy harvesting).

AAT: automatic antenna tuning, be sure that your antenna stays tuned

AAT will help to maximize performance in different situations

- External varicaps are used for trimming the matching circuit
 - 2 serial varicaps
 - 2 parallel varicaps
- Wide possibilities of tuning
- Internal DAC used for generating control voltage
- 2 independent 8-bit DAC
- Output voltage from 0 to 3.3 V

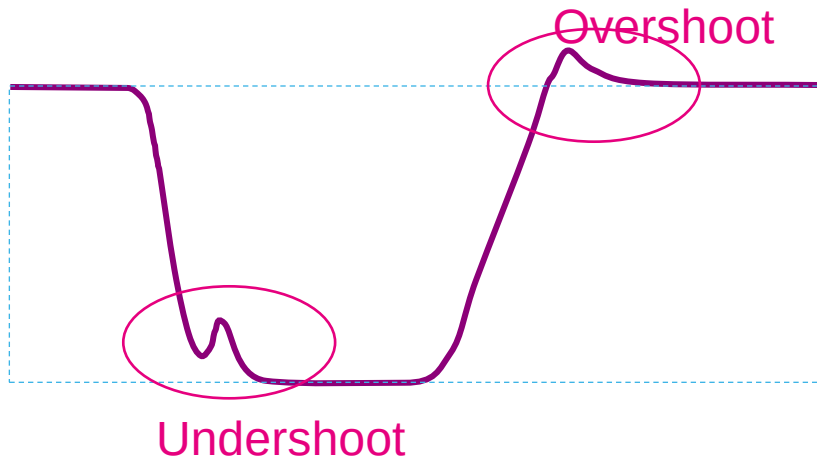


Active Wave Shaping



Fit wave shape requirements while offering good read range

Consider compliance of the system



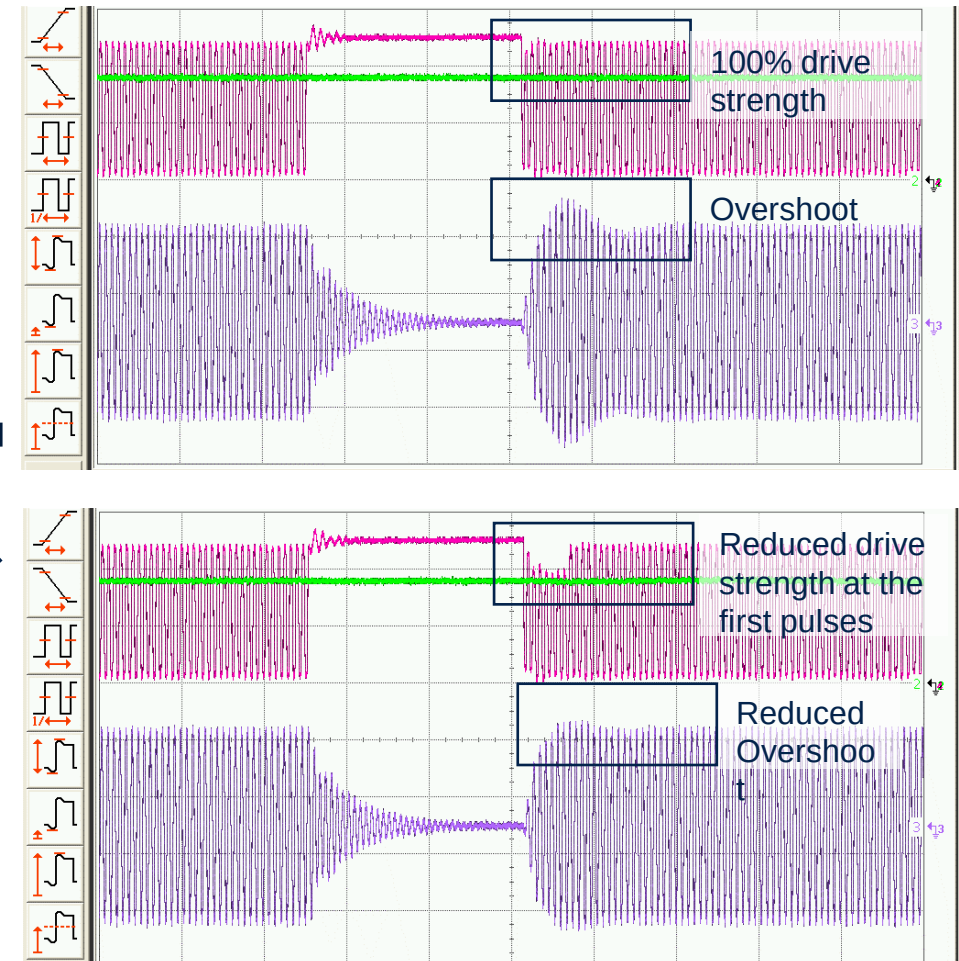
Option1: Change antenna tuning

- Hardware change: time consuming
- Reduce Q-factor: decrease operating distance

Option2: Use AWS

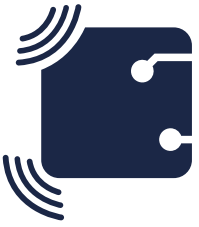
- Register setting: Low effort
- No tuning change: same range performance

AWS

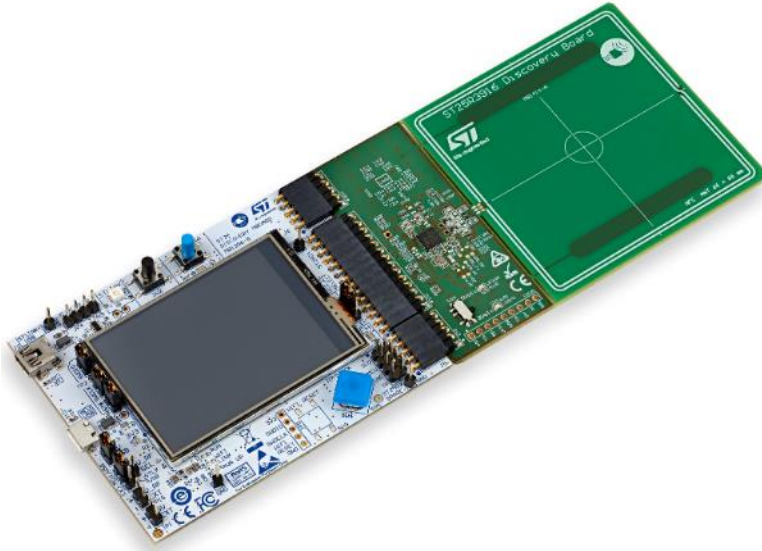


Support tools demo





ST25R3916B evaluation boards



ST25R3916B-DISCO

ST25R3916B discovery kit

- **ST25R3916B** high perf NFC universal device and EMVCo® reader
- 66x66mm 2 turns antenna etched on PCB
- STM32L476 ULP 32-bit MCU
- Micro-USB connector
- Additional UART/I²C host interfaces, as well as NFC SPI and JTAG/SWD points



X-NUCLEO-NFC08A1



ST25R3916B Nucleo shield

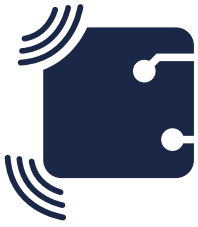
- **ST25R3916B** high perf NFC universal device and EMVCo® reader
- 47x34mm 4 turns antenna etched on PCB
- Compatible with STM32 Nucleo boards
- Equipped with Arduino® UNO R3 connector



ST25R3916B-EMVCO

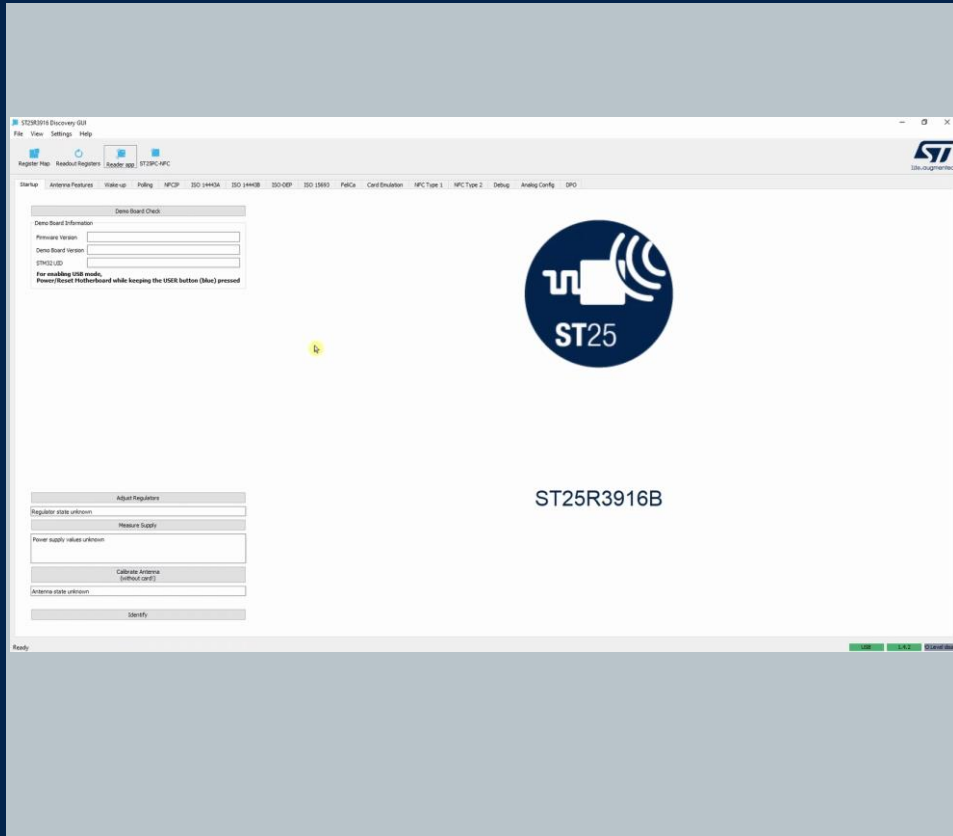
ST25R3916B EMVCo® kit

- **ST25R3916B** high perf NFC universal device and EMVCo® reader
- 73x65mm 2 turns antenna etched on PCB
- STM32L476 ULP 32-bit MCU
- Micro-USB connector
- Comprehensive device test environment (DTE) for EMVCo® level 1 FW control

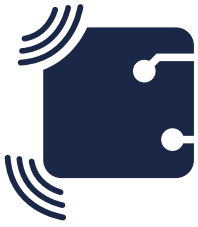


ST25R PC GUI software

ST NFC reader “Discovery” software suite



- Supports several RF protocols (ISO14443-A/NFC-A, ISO14443-B/NFC-B, FeliCa/NFC-F, and ISO15693/NFC-V)
- Active P2P (peer to peer) according to ISO18092, including SNEP
- Card emulation in NFC-A (106kbps) and NFC-F (212 and 424kbps)
- Wakeup feature, analog configuration, and register access of ST25R3916
- Support of automatic antenna tuning (AAT) & dynamic power output (DPO)
- Access to all ST25 tag features thanks to ST25PC NFC SW ([STSW-ST25PC001](#))
- ST25R3916 PC GUI SW ([STSW-ST25R010](#)) and ST25R3911B PC GUI SW ([STSW-ST25R001](#))



Presenting the demo



Demo Board Check

Demo Board Information

Firmware Version Demo Board Version STM32 UID **For enabling USB mode,
Power/Reset Motherboard while keeping the USER button (blue) pressed**

ST25R3916B

Adjust Regulators

Regulator state unknown

Measure Supply

Power supply values unknown

Calibrate Antenna
(without card!)

Antenna state unknown

Identify

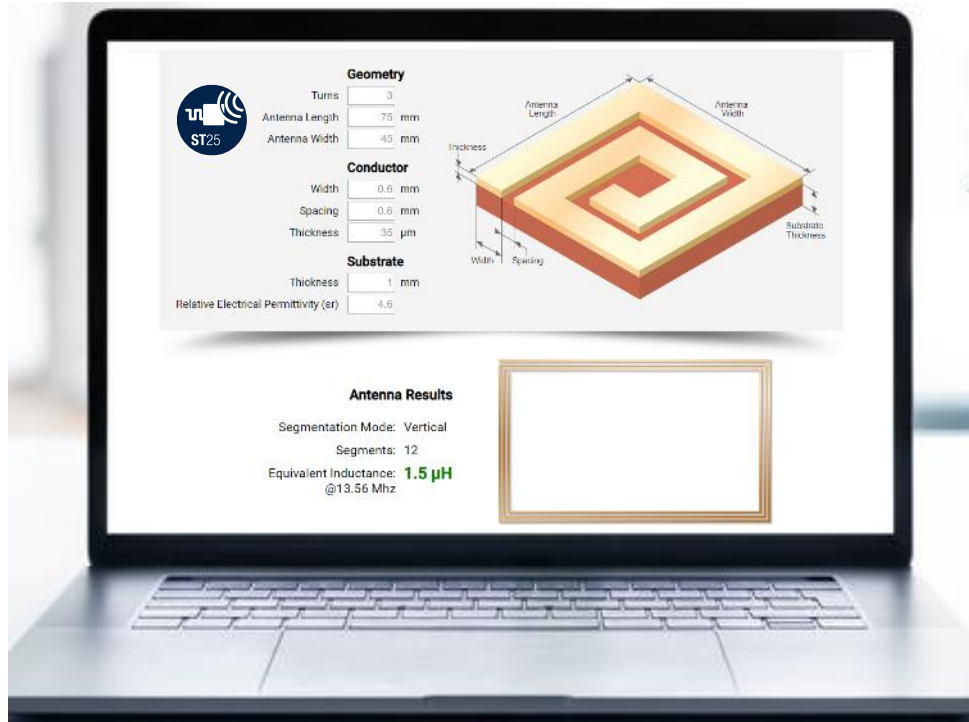
ST25 development tools



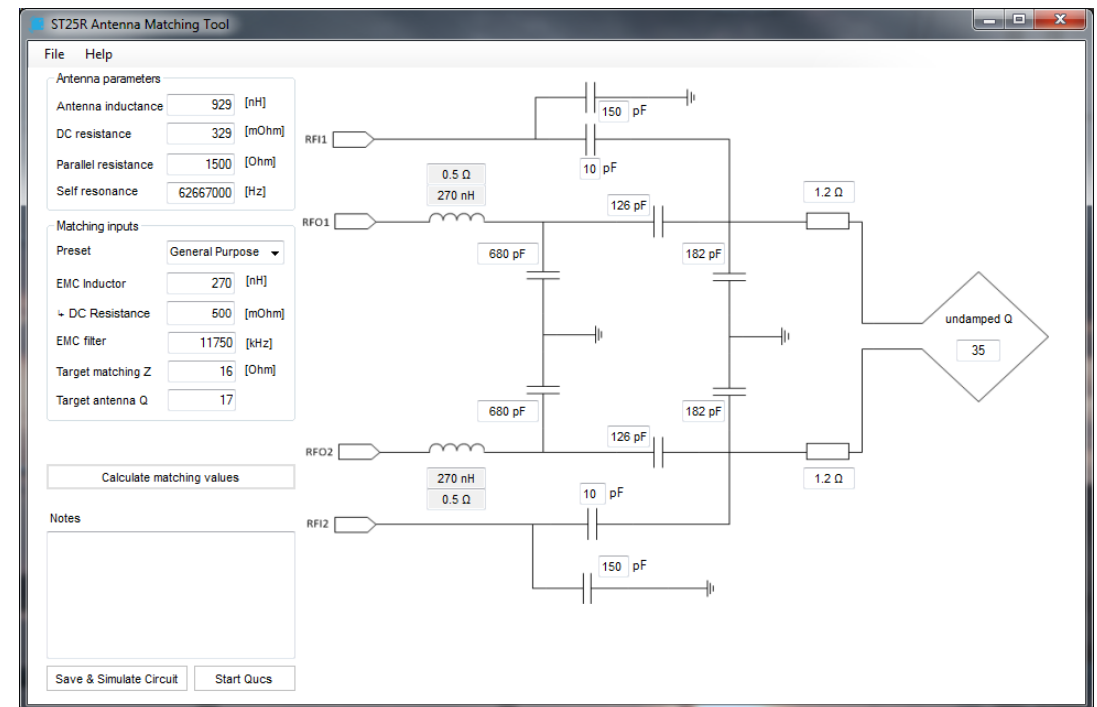
Antenna e-design and matching tools

Fast and easy prototyping

Antenna eDesign suite



ST25R antenna matching tool



ST25 eco-system DNA

Easy-to-use and customer-oriented



Antenna
e-design tool



Schematic,
BOM, Gerber



Discovery kit
STM32 based



STM32 Nucleo
Hardware ecosystem



Documentation



PC software tool
ST25 SDK



STM32Cube
SW ecosystem



Mobile apps
ST25 SDK

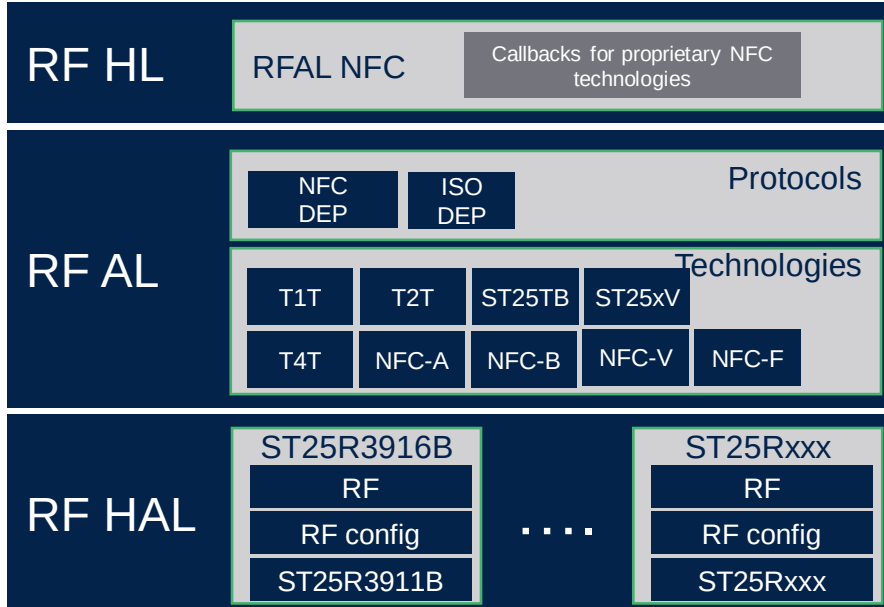


ST25R RFAL SW suite

ST NFC reader “RFAL” software suite



ST25RFAL

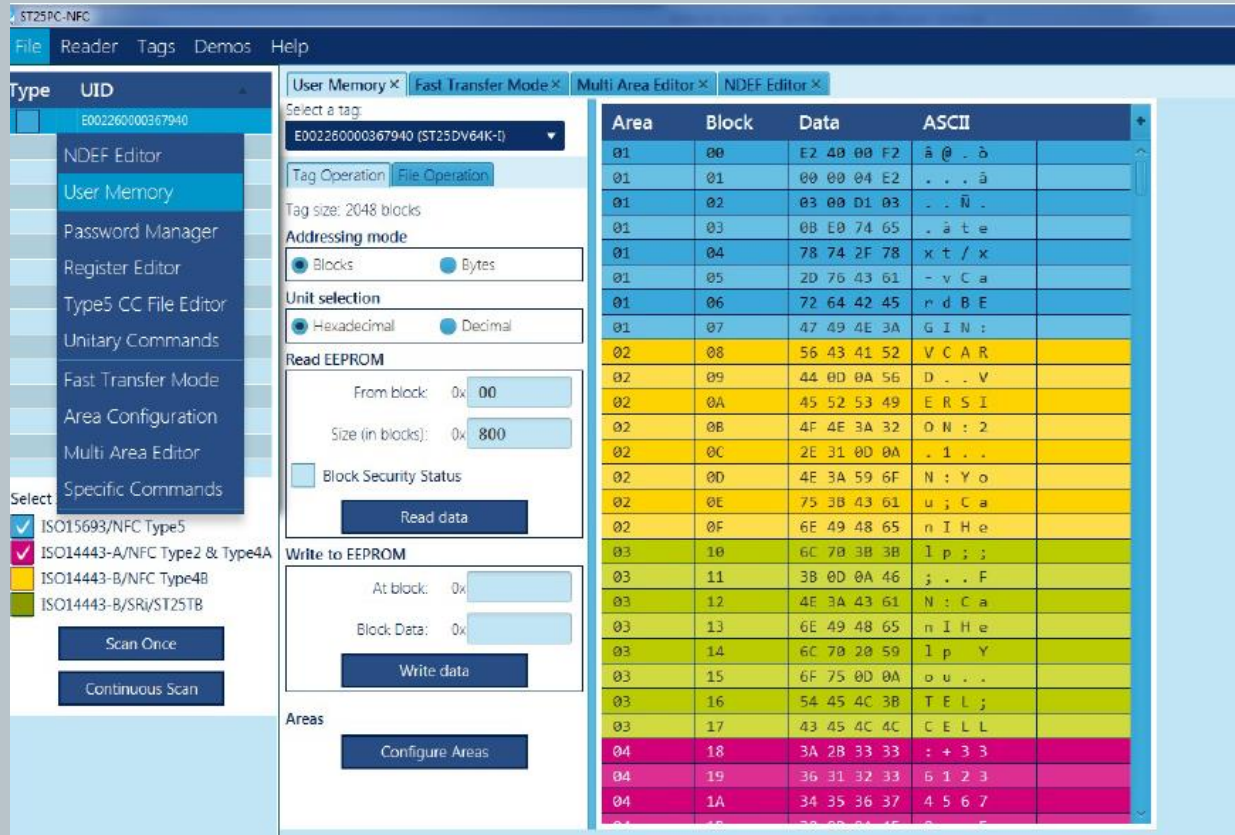


- Comprehensive device driver and middleware to build NFC enabled applications for reader devices based on ST25R NFC readers
- Written in pure ANSI C
- Straightforward portability across different platforms (MCU/RTOS/OS) with nonblocking API
- Compliant with main HF/NFC standards (NFC Forum, ISO)
- Source code examples available: embedded (STM32, STM8 device, SPC5 on request) and Linux® (Raspberry Pi)
- Easy callback function for proprietary NFC technologies on application layer like Apple Mfi (delivered under Mfi conditions) and other technologies
- ST25R3916 RFAL SW ([STSW-ST25RFAL002](#)) and ST25R3911B RFAL SW ([STSW-ST25RFAL01](#))



ST25 PC software

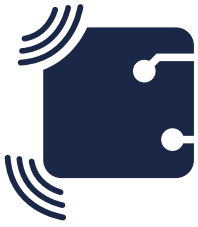
ST25 PC software for ISO15693, ISO14443-A/B & NFC readers



- Feature set support of ST25 NFC tags and dynamic tags
- PC SW for Windows
- Read/Write NDEF records on multiple tags
- Support of TruST25 digital signature feature
- Compatible with ST25R3916, ST25R3911B & CR95HF demo boards and industrial readers (FEIG)
- Fast transfer mode (FTM) demo with ST25DV-DISCOVERY board
- Free to use demo PC SW ([STSW-ST25PC001](#)) and open-source code ([STSW-ST25PC002](#))

ST25 readers mapping





ST25R readers - overview

Find the right reader for the right application

ST25R device	Output power	IWU	AWS	DPO	AAT	Focus applications
ST25R3912	1W	x		x		Accessory recognition, gaming...
ST25R3914	1W	x		x	x	After market identification
ST25R3915	1W	x		x		After market identification
ST25R3916B	1.6W	x	x	x	x	Payment, access control
ST25R3917B	1.6W	x	x	x		Payment, access control
ST25R3918	0.5W	x	x	x		Accessory recognition, gaming...
ST25R3920B	1.6W	x	x	x	x	Car access, bank card protection



AAT: Automatic antenna tuning

DPO: Dynamic power output

AWS: Active wave shaping

IWU: Inductive wakeup



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2nd Q&A Session





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Solutions for NFC/RFID tags & readers



ST25 SIMPLY MORE CONNECTED



Our technology starts with You



Find out more at www.st.com/st25

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